

Can We Trust AI in Healthcare? A Critical Look at Ethics, Reliability, and Accountability

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Abstract

As artificial intelligence (AI) becomes increasingly embedded in healthcare systems, it brings not only transformative potential but also profound ethical, legal, and social dilemmas. While proponents highlight AI's capacity to enhance diagnostic precision, streamline clinical workflows, and optimize resource distribution, its real-world implementation often reveals substantial risks. Chief among these are concerns over system opacity, data privacy, algorithmic bias, and the erosion of human accountability. Many AI systems operate as "black boxes," making it difficult for clinicians and patients to understand or challenge their outputs. At the same time, the integration of corporate-owned AI tools raises questions about the control and commodification of sensitive patient data. Moreover, machine learning models trained on biased or unrepresentative data can perpetuate and even amplify health disparities. Coupled with the lack of clear legal responsibility in the event of AI-related harm, these factors challenge the assumption that AI can be fully trusted in clinical decision-making. This paper critically evaluates these issues, arguing that until comprehensive safeguards are developed—including transparent design, equitable data practices, and robust accountability frameworks—AI in healthcare must remain a tool of cautious support rather than autonomous authority.

Introduction

Would you trust a machine to decide whether you have cancer? As artificial intelligence (AI) becomes increasingly integrated into healthcare, this question moves from speculation to reality. AI technologies, ranging from machine learning for genomics to computer vision in diagnostic imaging, now assist in interpreting vast amounts of medical data. For example, deep learning models like RED-CNN enhance CT scan quality, while AI-driven genomic analysis can predict drug responses in cancer patients, supporting more personalized care (Hirani et al., 2024).

Yet despite these advancements, AI raises serious concerns regarding fairness, transparency, and patient safety. Many AI systems function as "black boxes," making decisions without revealing their reasoning. This opacity is particularly troubling in high-stakes environments like hospitals, where physicians must justify clinical decisions. Xu and Shuttleworth (2024) emphasize that without explainability, trust in AI predictions is limited, and medical professionals may be unable to identify errors. Moreover, ethical concerns span the entire AI lifecycle—from biased datasets to unclear accountability and data misuse (Harishbhai Tilala et al., 2024).

These challenges lead to a central question: To what extent can we trust AI in healthcare, and what ethical, technical, and human-centered limitations undermine its reliability as a clinical decision-maker?

This essay critically addresses that question in five parts. First, it analyzes the opacity of AI systems and their impact on decision-making. Second, it explores algorithmic bias and discrimination in medical AI. Third, it examines the legal and ethical gaps in accountability. Fourth,

it discusses the erosion of human judgment and trust. Finally, it investigates privacy and consent issues related to health data.

Although AI offers useful tools, this essay argues that it should not be fully trusted in healthcare until it becomes more transparent, unbiased, and ethically accountable.

The Opacity of AI Systems

1. The Black Box Problem in AI Decision-Making

One of the most persistent barriers to fully trusting artificial intelligence (AI) in healthcare is the opacity of its decision-making processes—commonly referred to as the “black box” problem. This issue is particularly prominent in advanced machine learning techniques such as deep learning, where the internal mechanics of how an AI system arrives at a conclusion are often not transparent, even to the developers and clinicians who deploy them (Bjerring et al., 2025). Unlike traditional diagnostic tools, where the rationale behind a decision can typically be traced and validated, AI systems may deliver predictions or recommendations that lack clear interpretability. This creates a fundamental tension between high-performance outcomes and the ethical, legal, and clinical demand for explainability.

2. Ethical Implications of Opacity

The ethical implications of opacity are substantial. Xu and Shuttleworth (2024) argue that deploying black-box AI in healthcare contradicts the principle of “do no harm,” particularly when the system’s lack of transparency may lead to incorrect or biased decisions. Patients may suffer not only physical harm due to erroneous diagnoses or treatment plans but also psychological and financial harm if they are unable to understand or challenge the rationale behind a decision. The inability to fully explain AI outputs undermines informed consent and patient autonomy—two pillars of ethical medical practice.

3. Limitations of Explainable AI (XAI)

Furthermore, the current landscape of explainable artificial intelligence (XAI) has not adequately resolved these concerns. As Bjerring et al. (2025) highlight, many AI models provide only superficial explanations or post hoc interpretations that fail to capture the true complexity of the system's internal logic. These explanations often lack the granularity needed by clinicians to make confident decisions, especially in high-stakes situations. Moreover, the opacity of these systems can mask embedded biases in training data, potentially leading to unfair outcomes across different patient demographics without any clear indication of fault or accountability.

4. Clinicians’ Perspectives on AI Opacity

Nouis et al. (2025) conducted a qualitative study that explores how healthcare professionals in the UK perceive AI opacity. Their findings underscore that clinicians are not only wary of using tools

they cannot fully understand but also express concerns about how such tools integrate into their existing workflows and responsibilities. This research reveals that the opacity of AI is not simply a technical problem but a sociotechnical one, rooted in the interaction between users, systems, and institutional contexts.

The call for explainability is further supported by Freyer et al. (2024), who conducted a systematic review outlining the normative reasons behind the ethical requirement for explainability in AI-driven decision support systems (AI-DSS). These include ensuring accountability, enabling auditability, preserving clinician trust, and empowering patient participation. Freyer et al. (2024) also emphasize that trust in AI systems is inherently linked to their transparency and the ability to trace their outputs to clinically relevant reasoning.

Bias and Discrimination in Medical AI

1. The Ethical Urgency of Bias in AI

Bias and discrimination remain some of the most pressing ethical concerns surrounding the use of artificial intelligence in healthcare. These issues not only compromise diagnostic accuracy and patient outcomes but also risk exacerbating existing health disparities. A growing body of literature highlights the various sources and consequences of AI bias, offering both empirical evidence and strategies for mitigation.

2. Public Concerns and Perceptions

Public perceptions reflect deep concern over bias in AI-driven care. In a study surveying 600 Florida residents, respondents expressed greater comfort with AI in administrative roles (like appointment scheduling) than in clinical decision-making, citing fears of depersonalization and unequal treatment. Participants emphasized the need for transparency, accountability, and informed consent, especially among patients with lower healthcare self-efficacy. They also worried that biased algorithms could reinforce social inequalities unless AI is carefully regulated and designed with inclusivity in mind (Witkowski et al., 2024).

3. Clinical Concerns and Professional Insights

Clinicians echo similar worries. A UK-based qualitative study of healthcare professionals revealed real-world examples of AI underperforming for underrepresented populations. For instance, a biomedical scientist reported that an AI system consistently misclassified certain blood cell abnormalities in minority patients, raising safety concerns. A radiologist noted that while the AI excelled at detecting common pathologies, it often missed rare or atypical ones—likely due to non-representative training data. Developers and administrators acknowledged the need for continuous dataset updates, bias audits, and balanced training practices. However, they also stressed that biases often remain hidden until errors arise, especially in opaque or black-box systems (Nouis et al., 2025).

4. Empirical Evidence of Bias in Medical AI

Empirical studies reinforce these professional experiences. A systematic review of 11 cardiovascular AI models found that racial and ethnic bias was present in 82% of the studies, with algorithms often performing worse on Black, Hispanic, and Asian populations compared to White patients. Only five studies implemented mitigation strategies, and even then, the efforts were inconsistent across pre-processing, in-processing, and post-processing techniques. The review underscores the lack of standardization in addressing bias and highlights the urgent need for demographic-specific performance evaluation before clinical deployment (Cau et al., 2025).

5. Sources of AI Bias in Healthcare

The origins of AI bias in healthcare are both technical and systemic. (Ueda et al., 2024) identify two primary sources: data bias and algorithmic bias. Data bias stems from non-representative datasets, missing information, and reduced feature informativeness for marginalized groups. Examples include melanoma detection models underperforming on darker skin tones and cardiovascular tools failing to assess risk accurately in women due to male-dominated training data. Algorithmic bias, on the other hand, arises from design flaws such as using cost as a proxy for healthcare need—a practice that led to significantly lower predicted care needs for Black patients despite similar health statuses. Other issues include label bias (caused by inconsistent diagnostic labeling) and cohort bias (stemming from narrow training populations that ignore protected groups).

Together, these findings emphasize that AI does not operate in a vacuum; it reflects and can amplify the inequities of the systems it learns from. To ensure fairness, AI must be developed and deployed with careful attention to population diversity, real-world clinical complexity, and evolving social standards. Without systematic and sustained efforts to detect and correct bias, AI risks institutionalizing discrimination under the guise of innovation.

Accountability and Liability Issues

1. The Legal Complexity of Autonomous AI in Healthcare

As AI systems become more integrated into healthcare decision-making, questions of accountability and reliability grow increasingly urgent. Unlike traditional tools, AI systems often operate autonomously, learn from data over time, and produce outputs that are not always explainable, raising complex legal, ethical, and operational concerns. These issues are further complicated by fragmented global regulatory frameworks, unclear liability structures, and underdeveloped audit infrastructure.

2. Fragmented Global Regulatory Landscape

Globally, the regulation of medical AI primarily falls under the broader category of Software as a Medical Device (SaMD). However, many AI tools—especially administrative or wellness applications—escape regulatory oversight because they are not considered to directly influence clinical decisions. Even for high-risk AI applications, many jurisdictions rely on soft-law mechanisms like professional codes and voluntary standards, which offer flexibility but lack enforceability (Palaniappan et al., 2024). Regulatory approaches vary widely: the U.S. FDA is moving toward a Total Product Lifecycle (TPLC) model that emphasizes ongoing performance monitoring and machine learning best practices; the UK is developing a specialized AI-as-Medical-Device program; the EU’s forthcoming AI Act will impose strict requirements for high-risk systems; and Australia follows a risk-based model under the Therapeutic Goods Administration. Despite these efforts, a global harmonized framework remains lacking, which undermines trust and consistency in AI implementation (Palaniappan et al., 2024).

3. Ambiguity in Liability and Proposals for Legal Personality

The legal liability landscape is similarly ambiguous. Traditional tort law, which holds individuals liable for negligence or malpractice, struggles to adapt to autonomous systems. One controversial proposal has been to assign legal personality to AI-driven medical devices—effectively making the AI system itself liable for errors. Proponents argue this could streamline compensation and reduce the burden on physicians who are not fully in control of AI decisions. However, critics highlight several challenges: such devices cannot possess funds unless backed by their developers or institutions, and vicarious liability would still apply to hospitals and supervising physicians. Moreover, assigning intent or cognitive responsibility to machines remains philosophically and legally contentious (Eldakak et al., 2024). The debate ultimately centers on whether liability should follow control, benefit, or capacity to prevent harm, and no consensus has yet emerged.

4. Clinicians’ Reluctance and Fear of Accountability

Healthcare professionals themselves are acutely aware of these gaps. In a UK-based qualitative study, clinicians expressed reluctance to fully rely on AI tools, particularly when those tools function as "black boxes" with limited transparency. While some AI systems are valuable for detecting common conditions, clinicians noted frequent failures with atypical or minority cases. The fear of being held accountable for AI-driven mistakes—without the ability to audit or challenge decisions—undermines both adoption and trust (Nouis et al., 2025).

5. Gaps in Auditing and Oversight Infrastructure

Another critical issue is the lack of robust auditing and accountability infrastructure. A review of 435 existing AI audit tools revealed that most focus on performance evaluation, while tools for detecting harm, communicating audits, or ensuring transparency are far less developed. Many auditors report relying on ad hoc or self-built solutions due to the inadequacy of available tools.

This inconsistency makes it difficult to ensure that AI systems operate reliably and ethically in real-world healthcare environments (Ojewale et al., 2025).

Erosion of Doctor-Patient Trust and Human Judgement

The integration of Artificial Intelligence (AI) into healthcare has the potential to enhance efficiency, accuracy, and accessibility. However, this technological advancement is not without critical ethical and relational consequences. One of the most pressing concerns lies in its potential to erode the foundational elements of healthcare: the trust between doctor and patient and the irreplaceable value of human clinical judgment.

1. Patient Perceptions and the Threat to Human Connection

While many patients acknowledge the potential benefits of AI—such as improved accessibility, faster diagnostics, cost reduction, and even companionship in remote care—these advantages are not without caveats. A scoping review by Moy et al. (2024) found that negative attitudes typically arise when AI systems function without sufficient human oversight. Patients expressed discomfort with the idea of AI acting autonomously in critical care decisions, particularly due to the lack of human empathy, a key factor in building trust and delivering psychologically supportive care.

These concerns are heightened in emotionally sensitive or complex scenarios, such as cancer diagnoses or end-of-life decisions. Patients consistently preferred that AI play a supportive rather than substitutive role, and emphasized the importance of doctors being present to explain and interpret AI-driven recommendations. In this context, AI is viewed not as a replacement, but as an epistemic tool to assist—not replace—human decision-making (Frost et al., 2025).

2. The Inadequacy of Artificial Empathy

Efforts to make AI appear more “empathetic” have been met with skepticism. Frost et al. (2025) found that public trust is not easily won by mere simulations of empathy. People place intrinsic value on the physical presence, intuition, and moral judgment of healthcare professionals—qualities that go beyond algorithmic reasoning. Simply designing AI to mimic human empathy is not enough to maintain patient trust when human interactions are expected, especially during distressing circumstances.

This is particularly evident in psychiatry, where the complexity of human emotions, mental health assessments, and individualized treatment planning resists reduction to computational models. As noted by Sahoo et al. (2025), professionals in mental health settings were especially vocal about AI’s inability to replicate nuanced human communication and emotional interpretation, making it unsuitable as a substitute for direct human care.

3. Clinician Concerns: Devaluation of Expertise and Responsibility

From the healthcare provider’s perspective, AI’s growing role may inadvertently undermine professional autonomy and human judgment. Many clinicians fear becoming intermediaries—tasked with interpreting AI outputs rather than making independent decisions informed by experience and clinical intuition (Sahoo et al., 2025). In such a system, decision-making authority may shift from physicians to algorithms, diminishing both their responsibility and their role in fostering patient trust.

Additionally, providers raised ethical concerns about data privacy, systemic bias, and the ability of AI to handle complex or atypical clinical presentations. The prospect of AI systems making inappropriate or unexplainable decisions only deepens professional resistance and distrust, especially when these systems lack transparency or robust oversight mechanisms.

4. Trust and the Conditions for Acceptance

Despite these challenges, patient acceptance of AI varies depending on several influencing factors: familiarity with the tool, previous positive experiences, simplicity of use, and the presence of human supervision during AI implementation (Moy et al., 2024). However, no matter the demographic or clinical setting, trust is consistently tied to the assurance that human clinicians remain central to the care process.

Participants in Frost et al. (2025) study emphasized that healthcare workers are valued not only for their efficiency but also for their embodied presence and moral discernment. Their ability to act with “practical wisdom”—weighing facts, emotions, and ethical considerations—cannot be fully replicated by AI.

Privacy

1. Corporate Control and Patient Data Stewardship

The integration of artificial intelligence (AI) into healthcare systems presents significant challenges related to privacy, particularly concerning the control and use of sensitive patient data by private corporations. As AI tools increasingly process vast amounts of personal and medical information, concerns arise regarding the stewardship and governance of this data. To mitigate risks of misuse or unauthorized exploitation, transparent and accountable data governance frameworks that respect patient privacy and agency are essential, ensuring a balance between technological innovation and protection of individual rights (Williamson and Prybutok, 2024).

2. Risks of Re-Identification and Algorithmic Opacity

One of the primary privacy risks stems from the potential re-identification of patients through sophisticated techniques applied to pseudonymized or anonymized data. Despite efforts to anonymize datasets, advances in computational methods can sometimes reverse these protections,

thereby breaching confidentiality and undermining patient trust. Furthermore, the complexity and opacity of AI algorithms often obscure how patient data is used or shared within these systems, compounding concerns about transparency and informed consent (Williamson and Prybutok, 2024).

3. Differential Privacy and Technical Safeguards

From a technical standpoint, differential privacy offers promising approaches to balancing data utility with privacy protection. Recent empirical studies suggest that even modest levels of privacy-preserving techniques can significantly reduce the risk of patient data reconstruction while maintaining high AI model performance (Ziller et al., 2024). However, determining the appropriate "privacy budget"—the acceptable level of privacy risk versus model accuracy—requires careful context-specific assessment, as theoretical worst-case scenarios often overestimate practical risks.

4. Organizational Security and Infrastructure Vulnerabilities

Beyond technical safeguards, healthcare organizations face a range of data security vulnerabilities, including data breaches, inadequate encryption, and insufficient access controls. These risks threaten patient confidentiality and can result in financial and reputational harm to providers. Robust organizational measures—such as encryption protocols, strict access management, and breach response planning—are critical components in safeguarding health information throughout its lifecycle (Li et al., 2024).

5. Legal Frameworks and Regulatory Compliance

Compliance with legal and regulatory frameworks, such as the Health Insurance Portability and Accountability Act (HIPAA) in the United States, further reinforces patient data protection. These regulations establish standards for the use, disclosure, and security of health data and hold providers accountable for safeguarding privacy. As AI applications continue to evolve, adherence to these legal obligations must be integrated into AI development and deployment processes to maintain patient trust and uphold ethical standards in healthcare delivery (Li et al., 2024).

Conclusion

Despite the promises of speed, efficiency, and precision, the current implementation of artificial intelligence in healthcare is fraught with deep-rooted concerns that undermine its reliability and trustworthiness. The opacity of AI models makes it difficult for patients and clinicians to understand, challenge, or explain their outputs—raising concerns about transparency and informed consent. The handling of personal health data by private entities introduces serious privacy vulnerabilities, which neither technical fixes nor legal frameworks have fully resolved. Furthermore, the presence of bias in training data and the potential for discriminatory outcomes threaten to exacerbate existing health inequities. Most troubling is the absence of clear lines of accountability when AI systems cause harm, leaving patients vulnerable and legal systems

unprepared. As healthcare is a profoundly human-centered domain, decisions affecting lives and well-being must be guided by empathy, context, and ethical reasoning—not solely by opaque algorithms. Until AI systems meet rigorous standards of fairness, explainability, and accountability, they should serve only in assistive roles, always under human supervision.

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